

ON THE THEORY OF TONE REPRODUCTION, WITH A GRAPHIC METHOD FOR THE SOLUTION OF PROBLEMS.*

BY

L. A. JONES.

Physicist, Research Laboratory, Eastman Kodak Company.

THE problem of ascertaining the exact extent to which it is possible by the photographic process to produce a pictorial representation of an object which will, when viewed, excite in the mind of the observer the same subjective impression as that produced by the image formed on the retina when the object itself is observed has engaged the attention of workers in the field of photography for many years. There are many phases of the subject to be considered, and a complete treatment requires a careful analysis of the various factors upon which depend the operation of our visual perception of space and spatial relations. A complete analysis, therefore, leads not only into the realm of physical science, but also into those of psychology and philosophy. It is not the purpose of this paper to present such a complete treatment, but to deal with a single phase of the subject. A careful consideration of the various factors upon which depends our visual perception of space leads to the conclusion that brightness and brightness differences (*i.e.* contrast) are by far the most important factors which are reproducible by the photographic process. The form factor (including size, shape, position, definition, etc.) to which so much importance is usually attached and which is conditioned by the performance of the image forming system, depends for its effectiveness upon the proper rendition of brightness differences. The proper reproduction of brightness and brightness differences, therefore, is of preëminent importance, and it is with this brightness factor and the possibility of its correct reproduction by the photographic process that this paper will deal.

This problem of tone reproduction, as it is commonly called,

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has frequently been regarded as requiring only the correct reproduction in the picture of the actual physical brightness of the original, and of the actual contrast relations existing between various areas of the object. When we consider, however, that a given brightness or contrast may produce an entirely different subjective impression, depending upon the state of adaptation of the observer's eye, it is obvious that such a treatment of the subject cannot afford a complete solution of the problem. The problem may, in fact, be divided into two general parts, the one, which we may term the objective phase, dealing with the reproduction of the actual physical brightnesses and capable of being completely solved by purely physical methods; the other, the subjective phase, including a consideration of those factors which determine the nature of the subjective impressions produced by the action of given physical stimuli under various conditions, and requiring the use of psycho-physical methods and data for its adequate treatment.

Hurter and Driffield,¹ who may be considered as the founders of the science of photographic sensitometry, treated this subject to some extent, but confined themselves to a consideration of the objective phase. Lord Rayleigh² also has treated, although rather briefly, the objective phase of the subject. F. F. Renwick has made valuable contributions to the solution of this problem at various times, dealing in particular with the brightness reproduction possible when using the region of underexposure,³ and later presenting a more complete treatment of the entire subject both from the objective and subjective viewpoints.⁴

Recently A. W. Porter and R. E. Slade⁵ have published a paper on the subject in which they take exception to some of the conclusions reached by Hurter and Driffield, and outline the relations which they consider essential for the solution of the problem. They also deal only with the objective phase, the reproduction of the actual physical factors. Following this, two short papers of a somewhat controversial nature dealing with special cases of the tone reproduction problems have appeared, one by

¹ *Jour. Soc. Chem. Ind.*, May, 1890, and February, 1891.

² *Phil. Mag.*, vol. xxii, 1911, p. 734.

³ *Phot. Jour.*, vol. xxxvi, 1913, p. 127.

⁴ *Phot. Jour.*, vol. xl, 1916, p. 222.

⁵ *Phil. Mag.*, vol. xxxviii, 1919, p. 187.

Mr. F. F. Renwick⁶ and the other by Messrs. Porter and Slade.⁷ Finally Mr. Renwick⁸ has published a paper in which he gives in detail the method by which he obtained the results previously published (*loc. cit.*) relative to the objective phase of the subject. His conclusions are in substantial agreement with those arrived at in the following treatment of the subject although a quite different graphic construction was used.

During the past five years a large amount of work has been done in this laboratory on the theory of tone reproduction, and a satisfactory solution for the objective phase was obtained some two or three years ago. Application of this method was made to a special problem, the determination of the color coefficients of various selective photographic images, the results being published by the author in collaboration with R. B. Wilsey in the JOURNAL OF THE FRANKLIN INSTITUTE, Vol. 185 (1918), p. 231. While the precise method of procedure in the paper cited (see page 253) applies only to the special problem considered, which was not fundamentally tone reproduction, use was made of the basic principle which constitutes the method for the solution of the tone reproduction problem to be presented in this paper. The publication referred to is mentioned merely as an instance of a special problem in photographic research to which the more general method may advantageously be applied. Simultaneously with the work of tone reproduction, research in the field of visual sensitometry has been in progress, which has resulted in data applicable to the solution of the subjective phase of the more general reproduction problem. The results of this work have been published as communications from this laboratory by P. G. Nutting,⁹ Julian Blanchard,¹⁰ and Prentice Reeves.¹¹

While some particular details of the subjective data are still lacking, it is felt that sufficient information is now available to establish the general nature of the function necessary for the con-

⁶ *Phil. Mag.*, vol. xxxviii, 1919, p. 637.

⁷ *Phil. Mag.*, vol. xxxviii, 1919, p. 637.

⁸ *Phil. Mag.*, vol. xxxix, 1920, p. 151.

⁹ JOURNAL OF THE FRANKLIN INSTITUTE, vol. clxxxiii, 1917, p. 287; *Trans. Ill. Eng. Soc.*, vol. xi, 1916, p. 939.

¹⁰ *Phys. Rev.*, vol. xi, 1918, p. 81.

¹¹ *Jour. Opt. Soc. Amer.*, i, 1917, p. 148; *Astro. Phys. Jour.*, xlvii, 1918, p. 141; *Psychol. Rev.*, vol. xxv, 1918, p. 330.

version of objective reproduction into its subjective equivalent. In the meantime further work is being carried out in order to make possible a more precise quantitative evaluation of this function.

SIMPLIFYING ASSUMPTIONS.

In order to simplify the subject and bring it within the limits of a single paper, it will be necessary to make certain simplifying assumptions. Since the sensitivity of the photographic plate to radiation of different wave-lengths is in general radically different from that of the eye, objects in which color is present have not in general the same brightness when considered from the visual standpoint as when the evaluation is in terms of the photographic plate.

Let us, therefore, assume that all areas of the object considered are both visually and photographically non-selective (*i.e.*, colorless), and further that the optical system forming the image of the object on the negative material is likewise non-selective. Under such conditions values of visual brightness are directly proportional to photographic brightness. Thus this treatment will deal with brightness relations uncomplicated by any consideration of selectivity. Let us further assume that all photographic deposits considered are also non-selective in order that visual and photographic density values may also be considered as identical.

TERMINOLOGY AND SYMBOLS.

The problem to be treated involves so many operations, materials and inter-relations that it is of considerable importance at the outset to adopt a logical system of terminology and symbols. In order to obtain a satisfactory foundation, let us consider briefly the various steps in the complete cycle involved in obtaining a reproduction by the photographic process. An image of the object (*O*) must be formed on the negative material (*X*) by a suitable optical system. Under our first assumption it will be unnecessary to distinguish between the photographic and visual characteristics of the object. The characteristic of the object of interest is its brightness, while that of the image formed on the negative material is the illumination. The image illumination is transformed into an exposure by the action of the time factor. The exposed sensitive material is now developed, thus being transformed into a negative, *N*. The essential characteristic of the

negative is transmission or density. The negative is now used to form an image, either by contact or projection, on the positive material *Y*. The most useful specifications of this image is its illumination, for the determination of which it is necessary to know the illumination incident on the negative. The illumination of the image is transformed into an exposure by the factor of time. The exposed positive material is now transformed by development into a positive, *P*, of which it is necessary to know the density values. The positive becomes an objective reproduction, which in order to avoid confusion in the system of terminology we will refer to as the "material" reproduction, when it is illuminated, and is characterized by its brightness values. The character of the subjective reproduction resulting when this objective reproduction is viewed by an observer is dependent upon its brightness values and the retinal adaptation of the observer. Finally the evaluation of the subjective reproduction relative to the subjective impression due to the object, which we shall for sake of brevity call the "subjective object," involves a consideration of the brightness values of the object and the retinal adaptation existing when viewing that object.

The "things" involved therefore are:

	Symbol
1. Object	<i>O</i>
2. Negative sensitive material	<i>X</i>
3. Negative	<i>N</i>
4. Positive sensitive material	<i>Y</i>
5. Positive	<i>P</i>
6. Material Reproduction	<i>MR</i>
7. Subjective Reproduction	<i>SR</i>
8. Subjective Object	<i>SO</i>

The first six of these are objective or physical in nature, while the last two are subjective or perceptual. An evaluation of *O* in terms of *MR* gives a function which defines the extent to which actual objective reproduction of brightness is obtained under the limiting conditions assumed while an evaluation of *SO* in terms of *SR* gives the function required to determine the reproduction of the subjective impression.

The symbols as indicated for these "things" will be used as subscript letters attached to the symbol for the various physical factors applying to them.

The physical factors are:

	Symbol
Radiant Flux	F
Illumination	I
Brightness	B
Reflection Coefficient	R
Transmission Coefficient	T
Exposure	E
Time	t
Density	D

Some of the fundamental relations existing between these factors are:

$$E = I \cdot t,$$

$$D = \log \frac{I}{T}$$

$$B = I \cdot R \text{ or } I \cdot T.$$

If, for a given absorbing layer,

F = Incident Flux, and

F' = Transmitted Flux (or Reflected Flux),

$$T = \frac{F'}{F}$$

$$R = \frac{F'}{F}$$

As an example of the use of subscript letters according to this system:

E_n = Exposures given to negative material.

E_p = Exposure given to positive material.

D_n = Density of negative.

T_n = Transmission coefficient of negative.

B_o = Brightness of object.

I_n = Illumination incident on the negative (during printing).

Other particular values of the physical factors are indicated in a similar manner.

In Table I is given a schematic outline showing the various steps in the computation of the final result of a given tone reproduction problem. It will be noted that the process is divided into ten distinct steps represented by the Roman numerals *I* to *X*. The necessary factors required at each step are designated by the corresponding Arabic numerals. In the parenthesis designated by the letters *A*, *B*, *C*, etc., are statements either verbal or sym-

bolic of the transformation factors which convert the factor of one step into that of the next succeeding step. As the starting point, I , it will be noted that the "subjective object" is taken, by which is meant the subjective impression of brightness and brightness contrast excited in the mind of the observer. This impression, it is obvious, is the thing which it is desirable to reproduce rather than the actual physical brightness values which are measured by the usual physical methods. In practice, however, it is more convenient to take as the actual starting point of the operation the measured brightness values, B_o , of the object and to evaluate the subjective relation between B_o (II) and B_{so} (I) relative to that between B_{mr} (IX) and B_{sr} (X) after having obtained the necessary values of B_{mr} .

TABLE I.

- I. The Subjective Object (SO).
 1. Subjective brightness of object (B_{so}).
 - A. (Adaptation level of observer when viewing object, A_o)
 $B_{so} = f(A_o).$
- II. Object (O).
 2. Brightness (B_o).
 - B. (The constant of the image forming system, K_x .)
- III. Image on the X (negative) material.
 3. Illumination (I_x) = $B_o \cdot K_x$.
 - C. (Exposure time, t_x .)
- IV. Exposure on the X (negative) material.
 4. Exposure $E_x = (I_x \cdot t_x)$.
 - D. (Development, fixing, washing, drying, intensification, reduction, etc. Specified by $D_n = f(E_x).$)
- V. Negative (N).
 5. Density, D_n , or Transmission, T_n .
 - E. (Illumination, I_n .)
- VI. Image on the Y (positive) material.
 6. Illumination, $I_y = (T_n \cdot I_n)$.
 - F. (Exposure (printing) time, t_y .)
- VII. Exposure on the Y (positive) material.
 7. Exposure $E_y = (I_y \cdot t_y)$.
 - G. (Development, fixing, toning, etc. Specified by $D_p = f(E_y).$)
- VIII. Positive (P).
 8. Reflecting Power, R_p .
 - H. (Incident Illumination, I_p .)
- IX. The Material Reproduction (MR).
 9. Brightness (B_{mr}) = $(R_p \cdot I_p)$.
 - I. (Adaptation level of observer, A_{mr} . $B_{sr} = f(A_r).$)

X. The Subjective Reproduction (SR).10. Subjective Brightness (B_{sr}).

J. (The relation between B_{sr} and B_{so} determines the extent to which the desired subjective impression of tone is accomplished. $B_{so} = f(B_{so})$.)

I. The Subjective Object (SO).1. Subjective Brightness of Object (B_{so}).

THE NECESSARY DATA.

Before considering the method of obtaining the tone reproduction solution, it will be well to consider briefly the individual groups of data which must be utilized. These may be summarized as follows:

1. Relative to the Observer:

A_o Adaptation level of eye when observing the object.

2. Relative to the Object:

B_o Brightness values for various areas of the object.

3. Relative to Image Forming System:

K_g Factor converting B_o to I_g .

4. Relative to Negative Material:

$D = f(E)$ The H and D curve of the material.

5. Relative to Printing System:

I_n Illumination incident on negative during printing.

6. Relative to Positive Material:

$D = f(E)$ The H and D curve of the material.

7. Relative to Material (Objective) Reproduction:

I_{mr} Illumination on the print during observation.

8. Relative to the Observer:

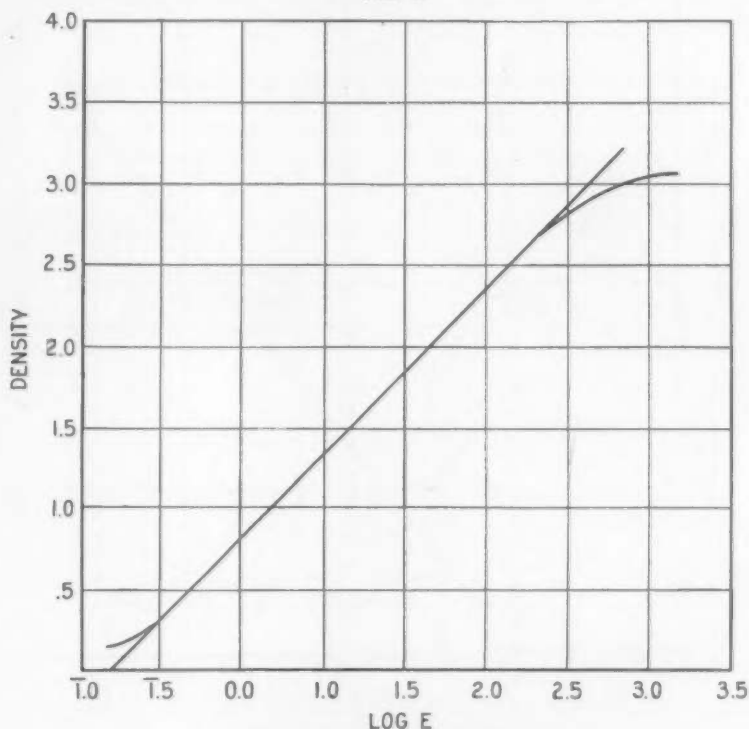
A_{mr} Adaptation level of the eye while observing the reproduction.

As a typical example of the data mentioned under No. 4, the curve in Fig. 1 is given. This is for a typical negative material of good quality, the ordinates being values of density (D) and the abscissæ the corresponding exposure plotted on a suitable logarithmic scale. In Fig. 2 is given a typical example of the same function ($D = f(E)$) for a positive material, in this case a photographic developing out paper. In case the positive is made on a transparent base, the curve will be of the same general type, but resembling more closely with respect to absolute values the curve shown in Fig. 1. In order to determine the relation between the objective and subjective reproduction, it is necessary in addition to the factors mentioned previously to use the sensibility function of the eye. In Fig. 3 are plotted the curves showing the retinal sensibility to brightness and brightness differences as a function of adaptation level. These curves are obtained from

data by Nutting and Blanchard, and constitute the most reliable information available relative to the subject.

Curve *A* is plotted with values of sensation (*S*) as ordinates against the logarithm of the field brightness (*B*). Curve *B* is the first derivative of *A*, and is plotted with values of $\frac{B}{\Delta B}$ as ordinates. This curve gives the contrast sensibility of the eye for the

FIG. 1.



Characteristic curve of negative material.

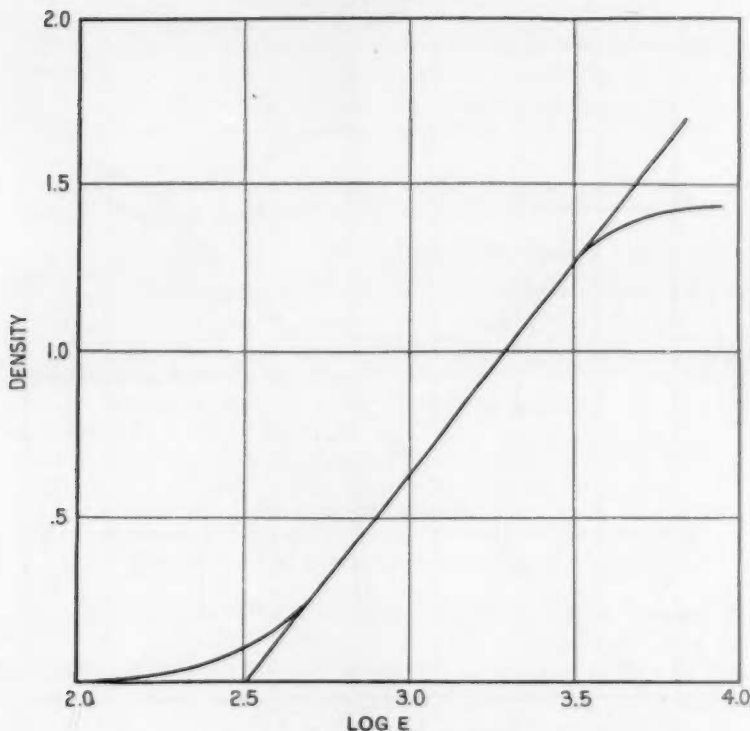
various values of *B* (plotted on a log scale). The relation between curves *A* and *B*, that the sensibility curve is the first derivative of the sensation (response), is a general one,¹² this being true for the corresponding curves of any physical instrument which indicates by some response (scale reading) the magnitude of the applied stimulus.

The data contained in the curves of the three figures (1, 2, and

¹² Nutting, P. G.: *Bull. Bur. Stds.*, vol. v, 1908, p. 266.

3) with the other values mentioned in the tabulation (p. 46) constitute the material from which a complete solution of the tone reproduction problem can be constructed. Having then made certain simplifying assumptions, established a satisfactory system of terminology and symbols, and considered briefly, in a general

FIG. 2.



Characteristic curve of positive material.

way, the problem and the necessary factors involved, let us proceed to the development of the graphic and analytical solutions.

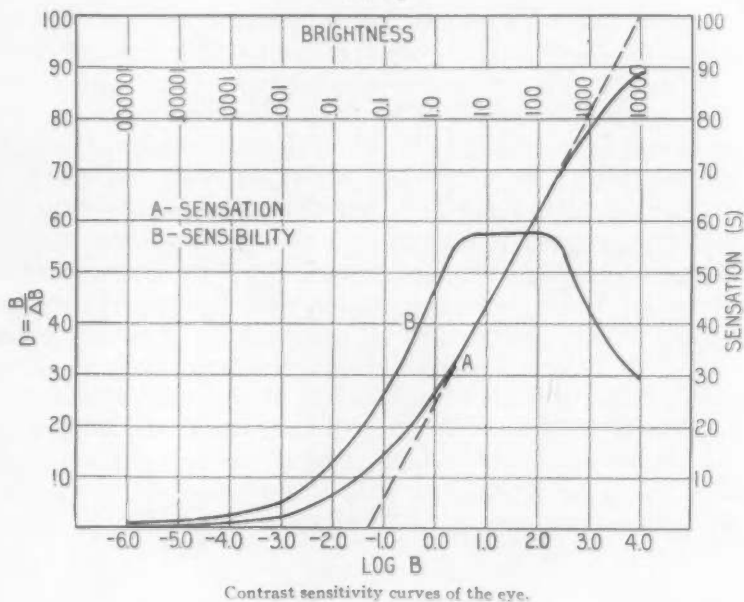
A GRAPHIC SOLUTION.

The graphic solution is presented in Fig. 4, the necessary curves being plotted in the four quadrants designated for reference as *I*, *II*, *III*, and *IV*. For convenience in plotting a double scale along the OX' axis, a separation between the upper and

lower quadrants is made, the two points designated by O being, in fact, identical.

Along the line OX is established a suitable logarithmic scale upon which may be plotted the logarithms of the brightness values for the various areas of the object. Let the points a_o , b_o and c_o represent three typical object-brightness (B_o) values, a being the lowest and c the highest which exist in the object considered. After having located on this scale ($\log B_o$) the various points

FIG. 3.



representing the brightness values of the object which it is desired to reproduce, the curve A is plotted directly under the $\log B_o$ scale, and properly placed in the horizontal direction relative to the points on the $\log B_o$ scale. This curve A is the characteristic H and D curve of the sensitive material on which the negative is made. It must be determined by sensitometric methods such that it is a true representation of the characteristic of the negative material when used under the conditions existing in the making of the negative, such conditions including those factors operative in transforming the latent into a real image as well as

sumption has been made that the deposits are non-selective, it follows that the color-coefficient is unity. It was assumed that the density values are determined under such conditions of illumination (diffuse or specular) that they are true values for the printing process used, contact or projection as the case may be. The unit interval on the $\log E_x$ scale must be equal to the unit used in establishing the $\log B_o$ and equal also to the unit interval of all log-scales used throughout the solution. The adjustment of the scale of $\log E_x$ along the line Y' , $Y'X$, depends upon the choice of the negative density value (D_n) by which it is desired to render a given object brightness. For instance, if it be desired to render point a_o , the lowest brightness in the object by a negative density of .20, the scale of $\log E_x$ is so adjusted that when curve A is plotted the perpendicular dropped from point a_o will cut the curve at a density value of .20.

The relation between the $\log B_o$ and the $\log E_x$ scale will now determine the exposure that must be given to the sensitive material on which the negative is to be made in order to obtain the rendition of the chosen brightness value by the desired negative density. In order to compute the exposure time, t_x , however, certain constants of the image forming system used in imaging the object on the sensitive material must be known.

Let the constant K_x be defined as that value by which a given value of object brightness, B_o must be multiplied in order to obtain the value of the illumination, I_x , incident on the negative material at the point where that particular area of object is imaged. K_x is, therefore, the value which satisfies the equation.

$$B_o \cdot K_x = I_x$$

$$\frac{E_x}{I_x} = t_x$$

$$\therefore t_x = \frac{E_x}{B_o \cdot K_x}, \text{ or}$$

$$\log t_x = \log E_x - (\log B_o + \log K_x)$$

In order to compute t_x (K_x being known), it is only necessary to read the values of $\log B_o$ and $\log E_x$ at the points where any line parallel to OY cuts the two scales and substitute in the equation.

The value of the constant K_x depends on several factors such

as diameter of stop, object and image distances, spectral transmission of the lens material, etc. The evaluation of K_s in terms of these various factors is somewhat complex, and will not be considered at this time.

Now it is convenient to regard the negative produced by the development of the exposed sensitive material (X) as a means of impressing a series of various exposures upon the sensitive material (Y) upon which the positive or print is to be made. The D_n scale must, therefore, be transformed into an inverse scale giving the relative values of the various illuminations which the negative will allow to act upon the material Y during the printing operation. Let I_n be the illumination incident upon the negative during printing, and I_y the illumination transmitted by the negative and therefore incident upon the positive material Y . Let T_n be the transmission coefficient of the negative. T is related to D (Density) by the equation:

$$D = \log \frac{I}{T}$$

and T is defined by the equation,

$$T = \frac{I_1}{I_0}$$

where I_0 = the incident illumination; I_1 = the transmitted illumination.

Hence:

$$\begin{aligned} I_y &= I_n \cdot T_n \\ \log I_y &= \log I_n + \log T_n \\ &= \log I_n - D_n \end{aligned}$$

Since I_n is a constant, differentiation gives $d \log I_y = -d D_n$. Hence, any interval on the $\log I_y$ scale is numerically equal to the corresponding interval on the D_n scale but of opposite sign. Now along the line OY' establish a log scale on which the value of I_y computed from the known values of D_n and some suitably chosen value of I_n may be plotted.

In quadrant III, plot curve B , the characteristic curve of the positive material (Y). This must be determined also under conditions which give the true characteristics of the material as used in making the print from the negative. The abscissæ values of this curve are the densities D'_p while the ordinates are the corresponding exposure values plotted on the scale of $\log E_y$ established in a suitable position on the line X' , $X'Y'$.

The scale of $\log E_y$ must be so adjusted along the line $X', X'Y'$ that the curve B when plotted will occupy the proper position in the vertical direction relative to the position of curve A . The proper position of curve B is determined by deciding by what value of D'_p it is desired to render some chosen value of D_n . For instance, let it be required to render the highest brightness, c_o , of the object by a just perceptible density on the positive. Assuming that the resulting print is to be viewed with an illumination such that the eye is operating in the region of maximum sensitiveness of brightness difference, a deposit differing in brightness by 2 per cent. from the background will be just perceptible. This is equivalent to a transmission or reflection coefficient of 98 per cent., which corresponds to a density of approximately .008. In order to fulfill the requirement, it will be necessary to adjust the scale of $\log E_y$ so that when the curve B is plotted a horizontal line drawn through the point where $D'_p = .008$ will cut curve A at c_n , which is also the point where a perpendicular dropped from c_o cuts curve A . The relation between the values of $\log E_y$ and $\log I_y$ at the points where any line parallel to $X'O$ cuts the two scales determines the exposure necessary to obtain the desired result. Thus:

$$\frac{E_y}{I_y} = t_y$$

or

$$\log t_y = \log E_y - \log I_y.$$

Substituting in this equation the values of I_y and E_y the value of t_y , the exposure time necessary may be determined.

It will be remembered that in order to establish the scale of $\log I_y$ a value of I_n was assumed. Since an arbitrary choice of this value, without previous knowledge of the speed of the positive material Y may lead to inconvenient or absurd values of t_y , it may be more logical to omit the establishment of the I_y scale and take the relative value of $\log E_y$ and D_n at the points where any horizontal line cuts the respective scales as a means of determining the exposure necessary in making the positive.

Thus:

$$\log t_y = \log E_y - \log I_y,$$

but,

$$\log I_y = \log I_n - D_n,$$

therefore,

$$\log t_y = \log E_y - \log I_n + D_n$$

The corresponding values of $\log E_v$ and D_n having been determined, two variables, I_n and t_v , remain, and by choosing a convenient value of t_v the value of I_n (the illumination incident on the negative during printing) necessary to satisfy the equation may be found.

Now let the points a_p , b_p , and c_p be located on curve B at the points where horizontal lines passing through the points a_n , b_n , and c_n intersect curve B , the points a_n , b_n , and c_n having been located at the intersections of curve A with perpendiculars dropped from points a_o , b_o , and c_o .

Along the line OX' lay off a log scale opposite in direction to the scale of $\log B_o$. The scale OX' is that from which the reflection coefficient (R_p) (or transmission coefficient T_p) of the various areas of the positive are determined. The position of this scale along the line OX' is determined by consideration of the values on the D'_p scale and the reflection or transmission coefficient of an area of the positive which has received no exposure. Let R_b (or T_b) be the reflection coefficient of such an area.

The values of D'_p are derived from the measurement of the reflection or transmission coefficient of the various areas of the positive relative to the reflection or transmission coefficients of an unexposed area of the positive material which has received, of course, the same development treatment as the other areas of the positive. These values, therefore, are of relative reflection (or transmission) coefficients. This method of measuring and specifying positive densities is preferable (especially in the case of positives to be viewed by reflected light) from the standpoint of practice, and is the usual procedure in the sensitometry of positive materials. But in order to find the brightness of the various areas of the positive, when observed under a given condition of illumination, it is necessary to know their absolute reflection coefficients. It is necessary, therefore, in order that the values of R_p (or T_p) read from the $\log R_p$ scale may be in absolute terms and suitable for the computation of resulting brightness values (B_{mr}) to lay off the values of this scale ($\log R_p$) so as to include the factor R_b (or T_b). The relation between the relative and absolute reflection (or transmission) coefficients is given by

$$R'_p \cdot R_b = R_p$$

or

$$\log R'_p + \log R_b = \log R_p$$

Where R'_p is defined by the equation

$$\begin{aligned}\log R'_p &= -D'_p \\ \therefore \log R_p &= +\log R_b - D'_p\end{aligned}$$

The scale on OX' must therefore be so constructed that at corresponding points on the $\log R_p$ and $\log D'_p$ scales the relation between the values of R_p and D'_p shall be as indicated in this equation.

This equation for the case of a positive material on a transparent base and intended to be observed by transmitted light becomes

$$\log T_p = +\log T_b - D'_p$$

In such cases it is sometimes more convenient to measure the values of D'_p in absolute terms, that is, they already include density of the unexposed area. The term T_b therefore becomes unity, making $\log T_b$ zero and the relation is expressed simply as,

$$\log T_p = -D'_p$$

Now the points a'_p , b'_p and c'_p are located on the $\log R_p$ scale at the intersections of the perpendiculars through the points a_p , b_p , and c_p , with the line OX' .

The values read from the scale at these points (a'_p , b'_p and c'_p) are the reflection (or transmission) coefficients of the areas on the positive by which the areas of the object represented by the points a_o , b_o and c_o are rendered.

The brightness (B) of a surface in terms of its reflection coefficient (R) and the incident illumination (I) is given by the relation,

$$\begin{aligned}B &= I \cdot R \\ \log B &= \log I + \log R\end{aligned}$$

Assuming then that the illumination on the positive during observation is I_p the brightness of any given area of the material (objective) reproduction is obtained by the relation

$$\log B_{mr} = \log R_p + \log I_p$$

The brightness of the areas a_p' , b_p' , and c_p' can, therefore, be computed for any assumed value of the illumination under which the positive is observed or the value of I_p can be computed for any assumed brightness for a given area. On the line OX' establish a log scale such that for corresponding points on this and the log

R_p scale (b_p' and b_{mr} , for instance) the values of $\log B_{mr}$ and $\log R_p$ will with the assumed value of I_p satisfy the equation

$$\log B_{mr} = \log R_p + \log I_p$$

Or, if it is desired that any given area of the positive be of the same brightness as the corresponding area of the object, the value of I_p necessary can be computed by the relation.

$$\begin{aligned}\log B_o &= \log B_{mr} = \log R_p + \log I_p \\ \log I_p &= \log B_o - \log R_p\end{aligned}$$

where $\log B_o$ and $\log R_p$ are the values for corresponding points of object and positive such as, for instance, b_o and b_p' . Since the relation between the reflection coefficients of the various areas of the positive may be entirely different from the relation between the brightness values of the corresponding areas of the object, it is in general possible to realize this equality of brightness condition for only one point on the positive. In practice, however, it is found that by proper adjustment of conditions it can be realized throughout a finite range, which in some cases is a considerable proportion of the entire tonal range of the object.

Now in considering the extent to which a given positive observed under given illumination reproduces the subjective impression caused by observation of the object itself, it is necessary to take into account the state of adaptation of the observer's eye not only while viewing the reproduction, but also when looking at the object. The curve which determines the relation between what we shall term the material reproduction and the subjective reproduction is, in the complete graphic solution, plotted in quadrant *IV*, and is referred to as the subjective relative-contrast function. In order to avoid confusion, it will be well for the present to omit consideration of this step in the problem and to complete the first solution for material reproduction, after which a more comprehensible exposition of the subjective phase will be possible.

In order to obtain the reproduction curve, it is necessary to transfer the points a_{mr} , b_{mr} , and c_{mr} to the OY axis. This can be done by simply rotating the $\log B_r$ scale about the point O until it coincides with the line OY , but since later a curve (the subjective relative contrast function) will occupy quadrant *IV* it will be better to make use of what may be termed a "dummy" curve (C) in *IV*, this being simply a line making an angle of 45° with the

line OX' and passing through O . Vertical lines through the points a_{mr} , b_{mr} , and c_{mr} intersecting curve D locate the points a_a , b_a , and c_a on this curve and horizontal lines through these intersections locate by their intersection with the line OY the position of the point a_{mr} , b_{mr} , and c_{mr} on this scale. Now the intersections of the horizontal lines through a_{mr} , b_{mr} , and c_{mr} (on line OY) with the vertical lines through a_o , b_o , and c_o determine the points a_x , b_x , and c_x which, when connected as shown, established the shape and position of the reproduction curve D (in I). This affords a graphic representation of the relation between the brightness factors of the object and those of the material reproduction (the illuminated positive). Now, though the point on the log B_{mr} scale (on line OY) where log B_{mr} is equal to the value of log B_o read at the point O (on line OX) draw a line parallel to OX . In the figure this is represented by the broken line $O'M$. This line is the absolute x axis of the reproduction curve D while the line $O'Y$ is the corresponding y axis, O' being the absolute origin. The line OX can be retained as the absolute axis by displacing the point of intersection of the dummy curve C (IV) in the proper direction along the X axis (line X' , O , X) by a distance equal to OO' . (In the case shown in Fig. 4 this point would be to the left of O at O'' .) In practice, however, such procedure usually results in the displacement of the reproduction curve, D , in the vertical direction by an inconvenient amount and location of the new position of the absolute axis as indicated is usually more convenient. In some cases this method may result in the point O' falling outside the limits of the available coördinates, in which case its location is best indicated by a dimension line carrying a numerical indication of its position relative to the apparent axis OX .

Now a straight line drawn through O' and making an angle of 45° to the line $O'M$ is the curve E , of absolute reproduction, this term being used to denote an exact reproduction in the material reproduction of both the brightness and contrast of the object. For any point on the reproduction curve, D , the magnitude of the departure from an exact reproduction of brightness is given by the length of the perpendicular line limited by the two curves D and E and drawn through the point considered; while the relation of the slopes (first derivatives) of the curves at corresponding points (b_a and b_x , for instance) determine the magni-

tude of the departure from exact reproduction of the contrast in the object at the point (b) considered.

The fact that the curve E is the curve for the exact reproduction of brightness follows from the construction which is such that for any point on this curve $\log B_o$ is equal to $\log B_{mr}$, thus satisfying the necessary condition that $B_o = B_{mr}$. The fact that a given brightness interval of the object is represented by the same brightness interval on the curve E indicates that the contrast of the object has been exactly reproduced. A straight line at 45° to the axis is the necessary condition for exact contrast reproduction, while for exact reproduction of brightness this line must pass through a certain specified point on the $\log B_{mr}$ scale.

From the position of the curve E relative to D (the reproduction curve), it is possible to determine quantitatively the magnitude of the departure from the condition of exact brightness reproduction. The most convenient method of expressing the magnitude of this departure is by use of the ratio of the brightness of the object to the corresponding brightness of the reproduction, which ratio will be designated by the symbol ΔB .

$$\Delta B = \frac{B_o}{B_{mr}}, \text{ or}$$

$$\log \Delta B = \log B_o - \log B_{mr}$$

Hence, if on the $\log B_{mr}$ scale (line OY) the values for any pair of corresponding points on the curves E and D (b_a and b_x , for instance) be read, ΔB can be determined. Since for any point on E , $\log B_o = \log B_{mr}$, the value of $\log B_{mr}$ can be substituted for $\log B_o$. For instance, consider the pair of corresponding points a_x and a_a

$$\begin{aligned} \log \Delta B &= \log B_o \text{ (at } a_o) - \log B_{mr} \text{ (at } a_x), \\ \text{and } \log B_o \text{ (at } a_o) &= \log B_{mr} \text{ (at } a_a), \\ \therefore \log \Delta B &= \log B_{mr} \text{ (at } a_a) - \log B_{mr} \text{ (at } a_x). \end{aligned}$$

It also follows since the length of the line $a_o a_x$ is given by

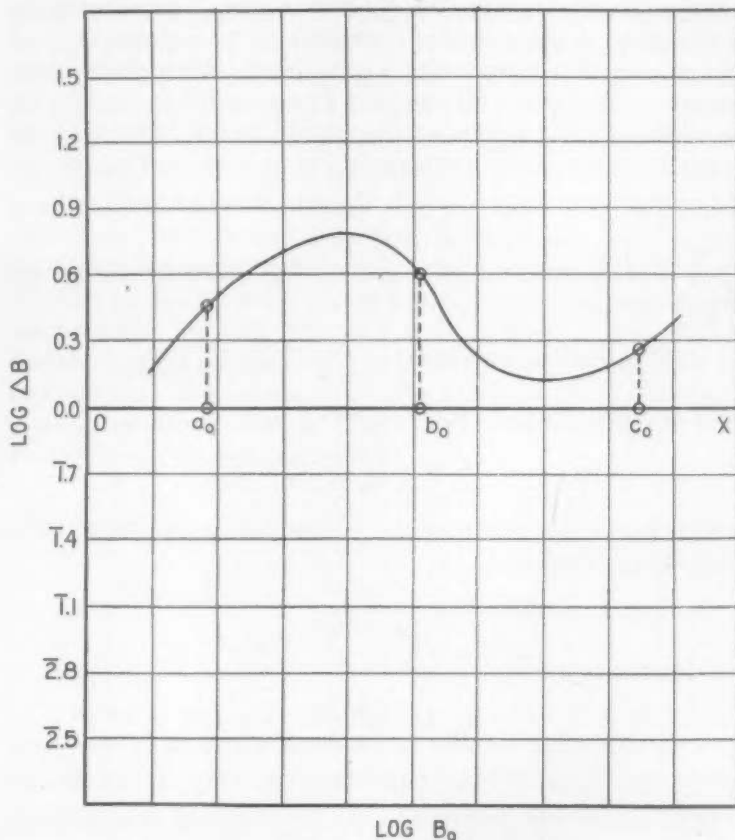
$$a_o a_x = \log B_{mr} \text{ (at } a_a) - \log B_{mr} \text{ (at } a_x).$$

that $\log \Delta B =$ the line $a_x a_a$.

Hence in order to plot the curve of brightness deviation in terms of its logarithm, it is only necessary to use as ordinates the lengths of the perpendiculars lying between the two curves D and E . For the sake of convenience and the avoidance of confusion due to a multiplicity of lines, this curve is not plotted in Fig. 4,

but is transferred to Fig. 5, the $\log B_o$ scale being a duplicate of that in Fig. 4 and the ordinates the values of $\log \Delta B$. The line OX passing through the point $\log \Delta B = 0$ ($\Delta B = 1.0$) is the line of no deviation, hence the axis of exact brightness reproduction. In plotting the curve of $\log \Delta B$ (Fig. 5) the sign of the

FIG. 5.



Curve of brightness deviations.

quantity $\log \Delta B$ must be considered. A point lying below the curve E (Fig. 4) will be represented in Fig. 5 by a point above the line OX and *vice versa*, a value of $\log \Delta B$ greater than 0.0, indicating that ΔB is greater than unity, and hence that B_o is greater than B_{mr} , while a value of $\log \Delta B$ less than 0.0 shows that B_o is less than B_{mr} .

ANALYTICAL CONSIDERATIONS.

Let us now apply the method of the differential calculus to the functions plotted in Fig. 4. Such treatment is especially useful in a determination of the contrast relations existing between the various elements. If at any point the ratio of some small increment ΔY (in the direction of the Y coördinate) to the corresponding increment ΔX (in the direction of the X coördinate) be computed the value obtained is a measure of the average slope of the curve over the distance (ΔX) considered. When these increments are reduced to the infinitesimal dy and dx the ratio gives the value of the slope at a point on the curve. Hence, if the slope or gradient be designated by the symbol G , it is defined by the expression $G = \frac{dy}{dx}$. Applying this then to the curves of Fig. 4, the following expressions are obtained.

The density gradient of curve A , the characteristic of the negative materials (X), is given by

$$G_n = \frac{d D_n}{d \log E_x}$$

Since $d \log E_x = d \log B_o$ this becomes

$$G_n = \frac{d D_n}{d \log B_o}$$

Likewise for B , the characteristic of the positive material (Y), the density gradient is,

$$G_p = \frac{D'_p}{d \log E_y}$$

While for the reproduction curve D

$$G_{mr} = \frac{d \log B_{mr}}{d \log B_o}$$

From the construction of the diagram (Fig. 4), it will be seen that

$$d \log E_y = -d D_n$$

and

$$d \log B_{mr} = -d D'_p = d \log R_p$$

$$\therefore G_n = - \frac{d \log E_y}{d \log B_o}$$

and

$$G_p = - \frac{d \log B_{mr}}{d \log E_y}$$

$$G_n \times G_p = - \frac{d \log E_y}{d \log B_o} \cdot - \frac{d \log B_{mr}}{d \log E_y} = \frac{d \log B_{mr}}{d \log B_o} = G_{mr}$$

$$\therefore G_n \cdot G_p = G_{mr}$$

Now for exact reproduction of contrast, it is necessary that $G_{mr} = 1.0$ and the condition fulfilling this requirement is that the product of the gradient (or slope) of the negative characteristic by the gradient of the positive characteristic shall be equal to unity. If $G_n \cdot G_p$ is greater than unity, a given brightness difference in the object ($d \log B_o$) is rendered in the positive by a brightness difference ($d \log B_{mr}$) greater than that existing in the object and hence the contrast is increased or the contrast scale is expanded. On the other hand, if $G_n \cdot G_p$ is less than unity it indicates that the brightness difference in the positive is less than the corresponding difference in the object ($d \log B_{mr} < d \log B_o$) and it follows that the contrast is decreased, that is, the contrast scale is compressed. Turning now to a consideration of the most convenient method of showing graphically the departure from exact reproduction of contrast, it is found that a curve showing the relation between B_o and the slope (G_{mr}) of the reproduction curve, D , fulfills all the requirements

$$G_e \text{ (Curve } E) = 1.0 \text{ by construction}$$

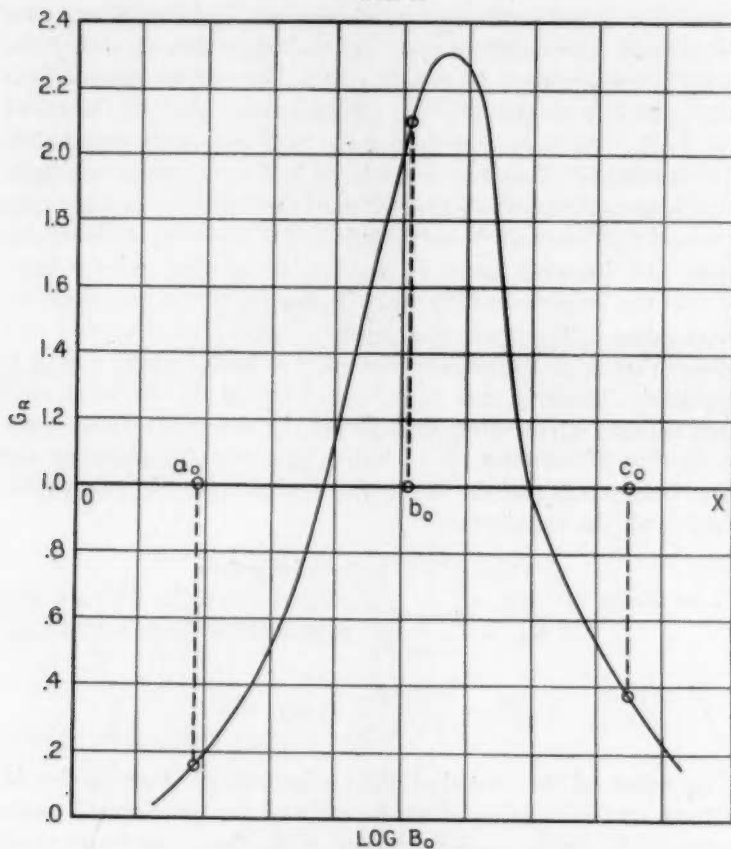
$$G_{mr} = \frac{d \log B_{mr}}{d \log B_o} \text{ (Curve } D)$$

$$\therefore \frac{G_{mr}}{G_e} = \frac{d \log B_{mr}}{d \log B_o} \text{ (Curve } D)$$

The value of the first derivative of curve D at any point is, therefore, numerically equal to the ratio of the gradient of curve D to that of E , and hence is a measure of the departure from exact contrast reproduction. In Fig. 6 is plotted the value of $\frac{d \log B_{mr}}{d \log B_o}$ (for curve D) as a function of B_o . The $\log B_o$ scale is again a duplicate of that in Fig. 4, and the ordinates are gradient values. The condition that $G_{mr} = 1.0$ is that for exact contrast reproduction, and hence the line OX ($G_{mr} = 1.0$) is the curve of exact contrast reproduction. Points lying above this line ($G_{mr} > 1.0$) indicate an increased contrast in the positive while points below ($G_{mr} < 1.0$) correspond to a diminished contrast. In case a given

difference of brightness ΔB_o in the object is rendered by a greater brightness difference ΔB_{mr} in the reproduction, the contrast scale is said to be expanded while the opposite condition is expressed as

FIG. 6.



Curve of contrast deviations.

a compression of the contrast scale. These conditions may be expressed mathematically, thus,

1. Increased contrast (Expansion of contrast scale).

$$G_{mr} > 1.0$$

2. Exact contrast reproduction (Normal contrast).

$$G_{mr} = 1.0$$

3. Diminished contrast (Compression of contrast scale).

$$G_{mr} < 1.0$$

In dealing with the straight line portion the gradient values are constant, and hence are direct measures of the contrast reproduction in such regions.

Thus far, no assumption has been made relative to the shape of the curves A and B , the treatment having been of a general nature. In practice it is found that both of these curves may, within the limits of observation, be straight lines over a considerable portion of their lengths. The tangent of the angle which such straight portion makes with the $\log E$ axis is designated by the symbol γ , and is equal to the gradient $\left(\frac{dD}{d \log E}\right)$ at any point on the straight line portion. The gradient being constant throughout such straight portions, it is convenient when dealing with tone reproduction by such portions alone to replace the values of gradient by those of gamma. The relation thus obtained is

$$\gamma_n \cdot \gamma_p = \gamma_{mr}$$

This it may be noted is the relation deduced by Porter and Slade (*loc. cit.*) which is a special case of the more general relation between the gradient values. The special case is, of course, simpler to deal with and is applicable when the entire reproduction can be obtained by use of the straight line portions, which frequently occurs when the positive is made to be viewed by transmitted light (transparencies, etc.). In the case of a positive to be viewed by reflected light, it is usually necessary to utilize the entire scale of the positive material, including the curved portions as well as the straight line, and in such cases the special relation ($\gamma_p \cdot \gamma_n = 1.0$, for exact contrast reproduction) is practically useless, it being necessary to employ the more general form, $G_p \cdot G_n = 1.0$.

For that brightness range of the object which can be rendered by the use of deposits in the negative and positive situated on the straight line portion of the characteristic curves of both materials, the above relation may be used. Stating this relation in words, we have for the most general case where the gradient is variable from point to point, that the product of the gradient of the negative by the gradient of the positive is equal to the gradient of the reproduction curve. This applies only, of course, to the gradient values determined at the corresponding points on the three curves, corresponding points being defined as any three points related to each other as the point b_n , b_p , and b_x , Fig. 4. Or, for the straight

line portions where gradient is constant and replaceable by gamma, we have that the product of the gamma of the negative by that of the positive is equal to that of the reproduction curve. It may be of interest to note at this point that this is precisely the same relation as that derived by the author in collaboration with Mr. Wilsey (*loc. cit.*), when applying this general method to the problem of measuring the color coefficient photographic deposits. And, since, for exact contrast reproduction, the slope (γ) of the reproduction curve must be unity, the necessary condition is expressed by

$$\gamma_n \cdot \gamma_p = 1.0$$

or

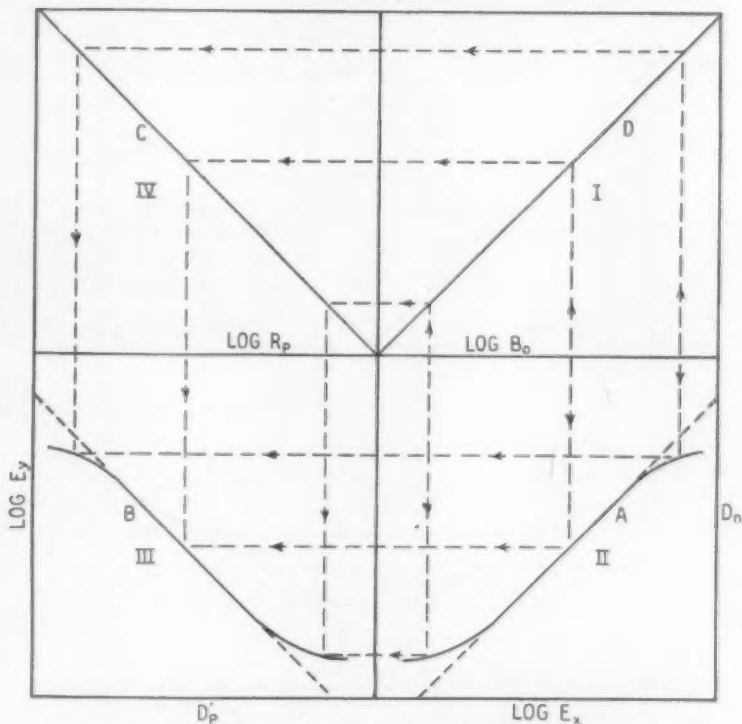
$$\gamma_n = \frac{1}{\gamma_p}$$

It should be borne in mind, however, that this relation holds only for the straight line portion of the curves, while the relation between the values of the differentials G_n , G_p , and G_{mr} is perfectly general and valid for any point regardless of the shape of the curves.

From a complete knowledge of the characteristics of the negative and positive materials, it is now possible to compute the quality of reproduction obtainable. It is also evident from the construction of Fig. 4 if only one of these functions is known, that that of the other can be obtained either graphically or by analytical methods provided an assumption as to the shape of the reproduction curve, D , be made. For instance, if exact reproduction of contrast be desired and a given negative material is to be used, it is evident that curve D must become a straight line having a gradient of unity ($G_{mr} = 1.0$). The shape of the positive characteristic necessary can now be determined. In Fig. 7 is shown the construction which determines this and the resulting shape in case the negative material has the characteristics indicated by curve A . Curve B is the necessary form of the function $D_D = f(\log E_D)$. It will be noted that this is of the same general shape as the "reciprocal curves" obtained by Mr. Renwick, which, however, cannot be realized in practice. While Mr. Renwick does not explain the method by which he arrived at the shape of his reciprocal curves, it was undoubtedly by a method somewhat similar to that presented in this paper; at least he arrives at the same general conclusion.

In a similar manner it is possible to determine the required shape of the negative characteristic ($D_n = f(\log E_x)$) in order to obtain any desired reproduction with a given positive material. The construction for this is shown in Fig. 8, exact reproduction of contrast being assumed with the positive material represented by the curve *B*. Figs. 7 and 8 serve to show the limita-

FIG. 7.



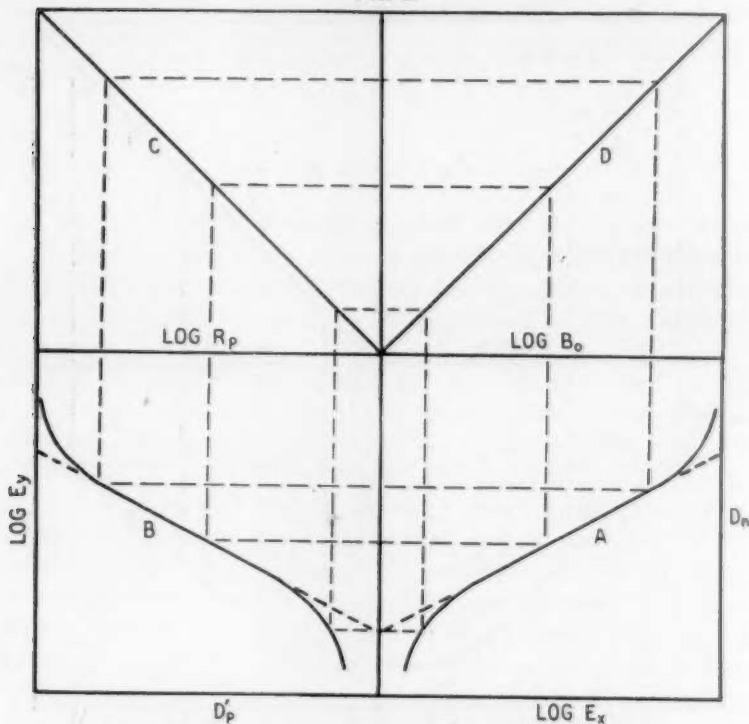
Characteristic of positive material required for exact reproduction.

tion of the photographic process in the exact reproduction of contrast, and since the method of construction must be obvious after the detailed discussion of Fig. 4, it will be unnecessary to discuss this further.

Another application of the method which may be mentioned is to the problem of finding the extent to which a negative must be developed in order to render any specified range of brightness values of the object on the straight line portion of the positive

material. In Fig. 9 assume that the range between a_o and b_o on the $\log B_o$ scale is to be rendered on the straight line portion of the positive material (Curve B). Locate the point a_p, b_p at the limits of the straight line portion of curve B , and find the positions of points a_n and b_n at the intersections of the vertical and horizontal dotted lines as shown. The line through a_n and b_n is the

FIG. 8.



Characteristic of negative material required for exact reproduction.

negative characteristic, A , required, the gradient of which is given by the relation,

$$G_n = \frac{d D_n}{d \log E_x}$$

or since we are dealing only with the straight line portions for which G_n is constant, $G_n = \gamma_n$

$$\gamma_n = \frac{\Delta D_n}{\Delta \log E_x} = \frac{\Delta D_n}{\Delta \log B_o}$$

Where

$$\Delta D_n = D_n \text{ (at } b_n) - D_n \text{ (at } a_n) = \text{Density Latitude (DL}_n\text{)}$$

$$\Delta \log E_x = \log E_x \text{ (at } b_n) - \log E_x \text{ (at } a_n) = \text{Exposure Latitude (EL}_n\text{)}$$

$$\Delta \log B_o = \log B_o \text{ (at } b_o) - \log B_o \text{ (at } a_o)$$

$$\Delta D_n = -\Delta \log E_y$$

Further

$$\gamma_p = \frac{\Delta D'_p}{\Delta \log E_y} = - \frac{\Delta \log R_p}{\Delta \log E_y}$$

and

$$\gamma_{mr} = \frac{\Delta \log B_{mr}}{\Delta \log B_o}$$

Where

$$\Delta D'_p = D'_p \text{ (at } a_p) - D'_p \text{ (at } b_p) = \text{Density Latitude (DL}_y\text{)}$$

$$\Delta \log E_y = \log E_y \text{ (at } a_p) - \log E_y \text{ (at } b_p) = \text{Exposure Latitude (EL}_y\text{)}$$

$$\Delta \log R_p = -\Delta D'_p$$

$$\Delta \log R_p = \Delta \log B_{mr}$$

As shown previously,

$$\gamma_n \cdot \gamma_p = \gamma_{mr}$$

and for exact reproduction of contrast,

$$\gamma_{mr} = 1.0$$

Now from the above relations, it can be shown in the case plotted in Fig. 9 that

$$\gamma_n = \frac{\Delta \log E_y}{\Delta \log B_o} = \frac{\text{Exposure Latitude of Positive Material.}}{\text{Brightness Range Assumed.}}$$

and that

$$\gamma_{mr} = \frac{\Delta D'_p}{\Delta \log B_o} = \frac{\text{Density Latitude of Positive Material.}}{\text{Brightness Range Assumed.}}$$

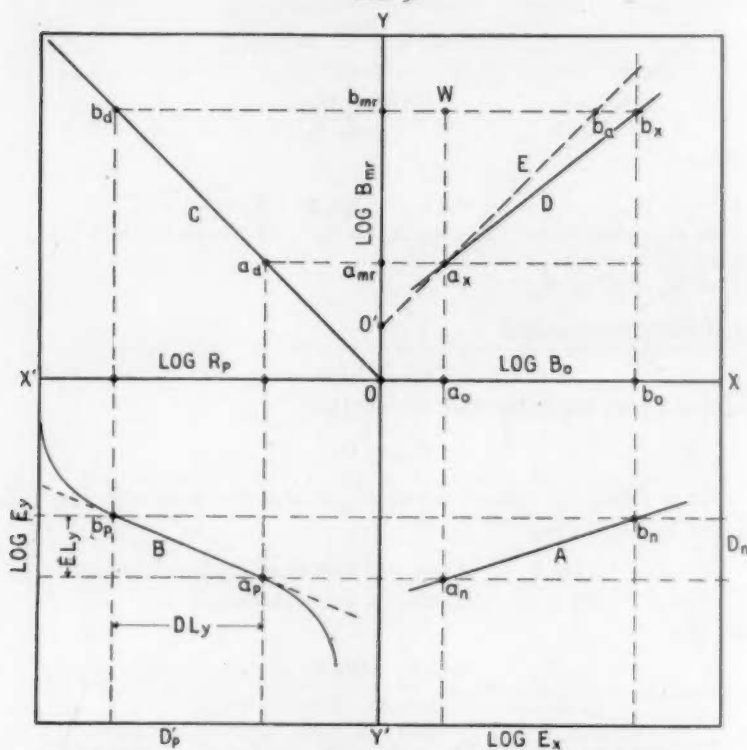
Also that in general the maximum brightness range of object that can be exactly reproduced is given by $\Delta \log B_o = \Delta D'_p$ (Density Latitude of Positive Material).

These statements assume that the exposure latitude of the negative material is equal to or greater than that of the positive material.

The writer wishes at this point to call attention to the fact that this conclusion, $\Delta \log B_o$ (Max. exactly reproducible) = $\Delta D'_p$, is the same as that drawn by Porter and Slade

(*loc. cit.*) in their recent publication, in which they state that such conclusion is at variance with that reached by the present author, who, in collaboration with Mees and Nutting, published some time ago a paper entitled "The Sensitometry of Photographic Papers" (*Phot. Jour.*, Vol. n. s., 38 (1914) p. 342). This paper, as indicated by its title, dealt with methods for the determination of the

FIG. 9.



Determination of the negative contrast required for specified results.

sensitometric constants of photographic printing papers, and did not pretend to deal with the problem of tone reproduction. A comparison of the characteristics of a negative material and a printing paper was made only for the sake of emphasizing the marked difference between them. Quoting the sentences in question we state: "But the maximum values of latitude met with in papers is of the order of .75 in log E units, or 1 to 5.6 in exposure

units. The paper then will only render correctly a range of brightness ranging from 1 to 5.6; hence it is obvious that in order to render a *negative* having a range any wider than 1 to 5.6 in *transmission* it is necessary to utilize portions of the characteristic curve lying outside the 'latitude' of the paper and thus depart from correct proportional rendering." It may be possible that the use of the word "brightness" in the first phrase (which was used as specifying the intensity factor of the light permitted by the negative to act upon the paper during printing) may have been misinterpreted. In the next sentence, however, it is specifically stated that a negative "having a range wider than 1 to 5.6 in *transmission*" was being considered. The word "transmission" should certainly have corrected any misapprehension arising from our use of the word "brightness." Moreover, in the following sentence of our paper, it is again clearly stated that the point under consideration is the "range of densities in the negative that can be correctly rendered," with no mention of the object brightness represented by such negative densities.

Returning again to further consideration of Fig. 9, it is evident if the range of brightness values ($\Delta \log B_o$) is greater than the density latitude (DL_y) of the positive material being used that it will be impossible to reproduce exactly the contrast of the object over the entire range, $\Delta \log B_o$.

However, it is possible to compress this tone scale of the object so that it can be proportionately rendered on the straight line of the positive material. Proportional rendering is defined by the relation, $d \log B_o = k \cdot d \log B_{mr}$, where k is a constant known as the proportionality factor.

$$k = \frac{d \log B_o}{d \log B_{mr}} = G_{mr}$$

or

$$k = \gamma_{mr} \text{ (For straight line portion).}$$

γ_{mr} may, therefore, be considered as the proportionality factor indicating the magnitude of the expansion or compression of the contrast scale. Thus, if $\gamma_{mr} = .5$ it indicates that a given brightness interval of the object has been compressed to one-half of its value in the reproduction, that is, the contrast has been decreased by one-half. The same applies to an expansion of the scale resulting in an increased contrast in the reproduction, the value

of γ_{mr} being again the proportionality factor. This, then, may be defined as that value by which a given brightness interval in the object must be multiplied in order to obtain the corresponding brightness interval in the reproduction.

It is evident from an inspection of Fig. 9 that the range of object brightness values which can be reproduced in correct proportion is limited only by the exposure latitude (EL_x) of the negative material. Since a negative material may have a latitude of 1 to 128 (in exposure units) or even more, it is evident that the great majority of objects may be rendered proportionally on the straight line portion, for measurements show that the contrast range seldom exceeds this value (1 to 128).

The applications of the methods outlined to practical problems are too numerous to mention in detail at this time. However, a specific example, Fig. 10, is shown. This illustrates the reproduction resulting from the use of obtainable negative and positive materials, the characteristic curves of which have been carefully determined by sensitometric measurements in this laboratory. The data, essential to this discussion, relative to the materials are as follows:

The negative material:

$$EL_x \text{ (Exposure latitude)} > 2.0 \text{ (in log } E).$$

$$\log i = \bar{2}.5$$

$$\gamma_{\infty} > 1.00$$

The positive material (a developing-out paper).

E_s	(Standard Exposure)	= 100,000 (m.c.s.)
γ_{∞}		= 1.34
EL_y	(Exposure Latitude)	= .77 (in log E)
ES_y	(Exposure Scale)	= 1.50 (in log E).
DL_y	(Density Latitude)	= 1.06
DS_y	(Density Scale)	= 1.43
D_{max}	(Maximum Density)	= 1.45
R_b	(Reflecting power of fog strip)	= .80

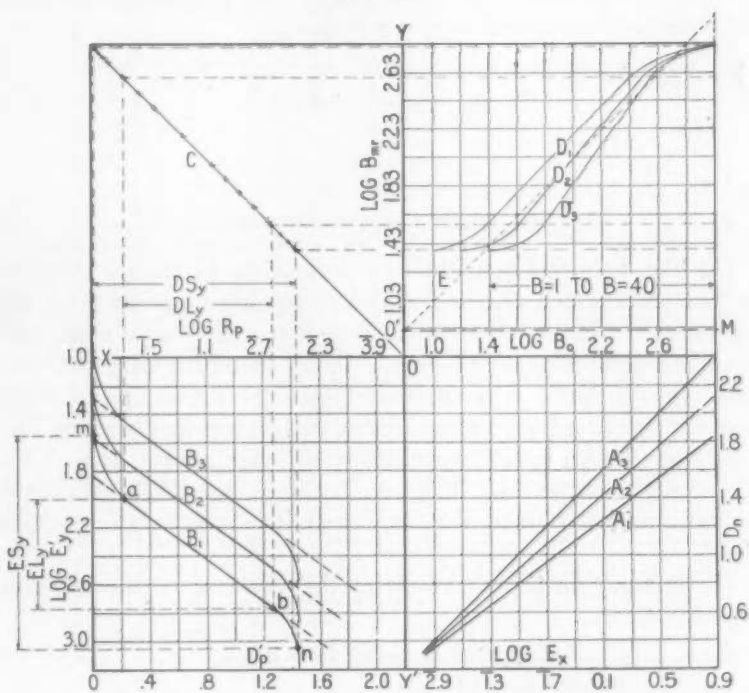
Let it be assumed that the highest object brightness is 1000 (apparent meter candles), and that this brightness is to be rendered in the positive by a just perceptible density ($D = .01$ (approximately), $R_p' = .02$).

In order to illustrate the effect of the contrast (γ) to which the negative is developed upon the quality of the resulting repro-

duction, three different negative characteristics are plotted differing only in the value of gamma. The gamma values chosen are for A_1 , $\gamma_n = .75$; for A_2 , $\gamma_n = .87$; and for A_3 , $\gamma_n = 1.00$.

The three curves B_1 , B_2 , and B_3 represent the characteristic of the positive material, these being exactly similar in shape but plotted in different position so that the condition of a just per-

FIG. 10.



Objective reproduction obtainable with specified materials and conditions.

ceptible density for the highest light may be fulfilled in all cases. The values on scale of $\log E'_y$ are arbitrary and a constant which differs for each of the curves B_1 , B_2 , and B_3 is applied in order to determine the value of $\log E_y$, from which the absolute value of printing exposure necessary may be computed, the relation being $\log E'_y + C = \log E_y$.

Curve C is the dummy used for transferring the points on the scale of $\log R_p$ to the $\log B_{mr}$ scale. Under certain conditions:

(previously specified) this curve becomes the actual subjective relative-contrast function. The points on this curve merely illustrate the way in which the construction points are located and apply to the construction of Curve D_1 . The curves D_1 , D_2 , and D_3 are reproduction curves obtained by the graphic solution as previously outlined, D_1 being that obtained by the use of A_1 and B_1 , etc. Since the positive material has a straight line portion, each reproduction curve is straight over part of its length. The gamma relation corresponds to the expression $\gamma_p \cdot \gamma_n = \gamma_{mr}$.

Therefore for D_1

$$\gamma_{mr} = 1.34 \cdot .750 = 1.01$$

for D_2

$$\gamma_{mr} = 1.34 \cdot .91 = 1.17$$

and for D_3

$$\gamma_{mr} = 1.34 \cdot 1.00 = 1.34$$

In Fig. 11 are plotted the first derivatives of the three reproduction curves D_1 , D_2 , and D_3 showing the magnitude of the departure from exact contrast rendering for each case, the straight line at $G_{mr} = 0$ (Fig. 11) being the curve of exact contrast reproduction. The curve E (Fig. 10) is the line of exact contrast reproduction, and so located as to represent also the best approximation to exact reproduction of brightness in case of the reproduction curve D_2 . The log B_{mr} scale is established on the assumption that I_p is of the magnitude indicated by the position of O' .

Now applying the equation for the value of t_x , the negative exposure time,

$$\log t_x = \log E_x - (\log B_o + \log K_x).$$

$$\text{Let } \log B_o = 2.6$$

$$\text{Then } \log E_x = 0.5$$

And assume as a value for the constant of the image forming system

$$K_x = .01, \log K_x = \bar{2}.0$$

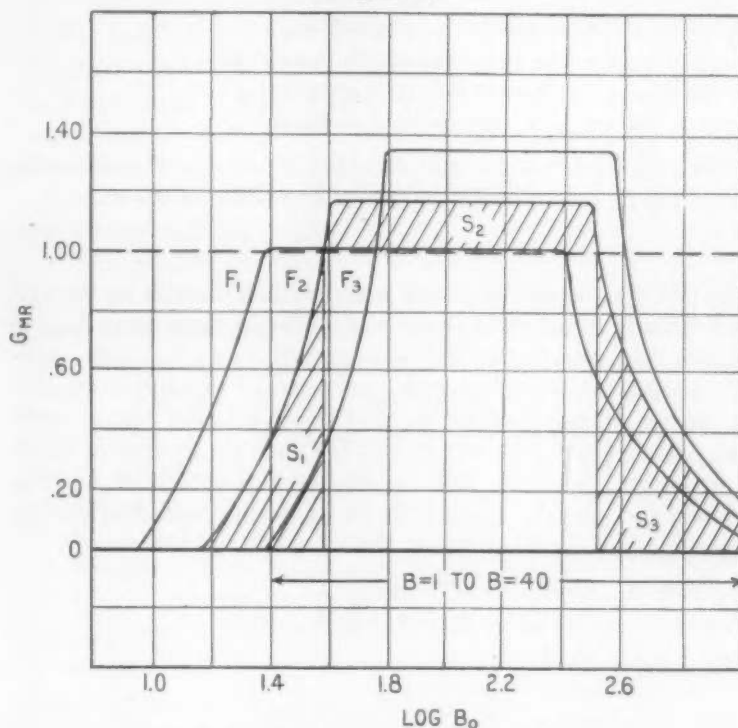
Solving we obtain

$$t_x = .0126 \text{ sec.}$$

This then is the time necessary in exposing the negative material.

The standard exposure, E_s , is defined (Jones, Mees, and Nutting, *l. c.*) as that exposure in meter-candle seconds which will just print through a density of 2.0. Since a density of 2.0 transmits .01 of the incident illumination it follows, in the present case, that an exposure of $.01 \times 100,000$ mcs. or 1000 mcs. will produce a just perceptible deposit on the positive material used.

FIG. 11.



Contrast deviations occurring in curves in Fig. 10.

Therefore at the point where

$$\log E'_y = 1.55, \log E_y = \log 1000 = 3.0$$

$$1.55 + C = 3.0$$

$$\therefore C_1 = 1.45$$

Likewise,

$$C_2 = 1.73$$

$$C_3 = 2.01$$

Now from the equation (p. 53) giving the printing exposure necessary,

$$\log t_y + \log I_n = \log E_y + D_n$$

$$\text{Where } D_n = 1.4$$

$$\log E_y = 2.0 + 1.45 = 3.45$$

$$\therefore \log t_y + \log I_n = 3.45 + 1.4 = 4.85$$

$$\text{or } t_y \cdot I_n = 70,800$$

$$\text{Hence, if } I_n = 1000 \text{ m. c.}$$

$$\text{For negative } A_n, t_y = 70.8 \text{ sec.}$$

$$\text{Inserting values of } C_2 + C_3 \text{ we obtain}$$

$$\text{For negative } A_n, t_y = 135 \text{ sec.}$$

$$\text{and for } A_n, T_y = 250 \text{ sec.}$$

Since R_b for the paper used is .8, $\log R_b = \bar{1}.90$. Therefore the scale of $\log R_p$ is so established that at any point on the scale,

$$\log R_p = \bar{1}.90 - D'_p$$

In this case the $\log B_{mr}$ scale is established directly on the OY axis. This is found to be more convenient in some cases than to establish this scale on the OX' axis as indicated previously, especially when it is desirable to make the curve of absolute reproduction occupy some particular position relative to the reproduction curve. In this case the curve E was drawn in the desired position, and from the point O' thus established the values of $\log B_{mr}$ necessary to make the curve E that of absolute reproduction was determined, the condition being that

$$\log B_{mr} (\text{at } O') = \log B_o (\text{at } O)$$

$$\log B_o (\text{at } O) = .8 = \log B_{mr} (\text{at } O').$$

Now from the equation,

$$\log I_p = \log B_{mr} - \log R_p$$

it is possible to find the value of I_p which is necessary to illuminate the positive to satisfy the relation between the $\log R_p$ and $\log B_{mr}$ scales as established.

The point on the $\log B_{mr}$ scale, for instance, corresponding to the point where $R_p = \bar{2}.3$ is $\log B_{mr} = 1.23$.

$$\therefore \log I_p = 1.23 - (\bar{2}.3)$$

$$2.93$$

$$I_p = 851. (\text{meter candles})$$

Now, with this illumination on the positive obtained by use of negative A_2 the material reproduction will be as indicated by the curve D_2 . In the region lying below the line E the areas will be less bright than the corresponding areas of the object, while in those above the contrary will be the case.

By decreasing the illumination the curve E can be made to coincide with the straight line portion of reproduction curve D_1 , and much better brightness reproduction obtained for that particular positive in the middle tones, although the departure in the highlights will be greater than for the curve D_2 with the illumination specified. In practice it is not necessary to obtain exact reproduction of brightness since the eye is an organ of variable sensibility, and a low actual brightness may excite the same sensation as one of high value if the adaptation level of the eye be of the proper value.

The location of the points m and n (Fig. 10) should be mentioned at this point. They are the points which limit the scale of the positive material, Y , and are located at the points on the curve of that material where the term $\frac{D_p}{d \log E_y}$ has a certain assumed value. The value assumed previously was .2 and occurs in the particular case of the printing paper used in this illustration at the points where $D = .01$ and $D = 1.44$. The points a and b are located at the extremities of the straight line portion, and their positions determine the values of the exposure latitude, EL_y , and the density latitude, DL_y . Since the gradient, G_{mr} , of the reproduction curve is equal to the product of G_n and G_p , it follows that the gradient of the reproduction curves at points corresponding to m and n will depend on the value of G_n , being greater for high values of G_n and *vice versa*. Practically, however, it may be considered that the scale of the reproduction curve is limited by the points corresponding to m and n as shown by the dotted construction, although in extreme cases exceptions to this rule may be necessary. Likewise the latitude of the reproduction curve may be considered as limited by the points corresponding to a and b . According to this it is evident from the construction that the following relation exists:

Defining as follows:

Object Latitude (OL) = Range of brightness values of the

object rendered on the straight line portion of the reproduction curve.

Object Scale (OS) = Range rendered on scale of reproduction curve.

$$OL = \frac{EL_y}{\gamma_n}$$

$$OS = \frac{ES_y}{\gamma_n}$$

Further if,

BL_{mr} = Brightness Latitude of the Reproduction.

BS_{mr} = Brightness Scale of the Reproduction.

Then:

$$BL_{mr} = DL_y$$

$$BS_{mr} = DS_y$$

It may be of interest to tabulate some of the actual values obtained in this curve.

TABLE II.

Negative	γ_n	γ_{mr}	OL	OS
A_1	.75	1.01	1.03	2.00
A_2	.87	1.17	.89	1.73
A_3	1.00	1.34	.77	1.50

Other useful relations may be desired, but since their importance depends upon the particular problem being dealt with, further consideration will not be given them at this time.

Turning now to the derivatives plotted in Fig. 11, their interpretation will be considered briefly. By integrating the area inclosed by each curve and the line of G_{mr} equal to zero, the mean value of gradient over the entire scale of object brightness rendered can be determined. These values are tabulated in column B (Table III).

TABLE III.

Negative	γ_n	B	C	D
A_1	.75	.71	.30	.25
A_2	.89	.83	.37	.32
A_3	1.00	.95	.45	.48

The mean deviation from the line of exact contrast reproduction can be obtained by integrating the partial areas as indicated by the shaded areas (S_1 , S_2 , and S_3) which apply to curve F_2 , adding the values thus obtained together and dividing by the

value of OS . The mean deviations thus determined are given in column C (Table III).

The deviations thus obtained apply, of course, to different ranges of brightness in the object. However, if it is desired to determine the mean deviation from $G=1.00$ for a given object having a fixed contrast, it is only necessary to limit the areas considered to the specified brightness range. For instance, let us determine the mean deviation for each reproduction of contrast for an object having a range of $B=1$ to $B=40$. The vertical line drawn through the point $\log B_o = 1.4$ now limits the areas which must be integrated in order to find the desired deviation. The values obtained for the three curves are given in column D (Table III). It is thus possible to determine the magnitude of the departure from exact reproduction of contrast of any desired range of object brightness values, when reproduced by means of negative and positive materials of known characteristics.

From a consideration of Fig. 10, a conclusion of considerable importance in practical work may be drawn. This relates to the problem of choosing the printing paper best adapted to a given negative or the inverse problem of making a negative of the proper quality for printing on a given printing paper. Considering Curve B , it will be seen that the useful portion is limited by the points m and n . On portions of the curve lying outside of these limits the gradient $\frac{d D'_p}{d \log E_y}$ is so small as to be practically useless in the reproduction of differences in E_y . Now in practice it is, in the great majority of cases, desirable to render the deepest shadow (lowest brightness) of the object by the maximum useful density of the positive material, and the highest light (greatest brightness) by the minimum useful density of the printing material, and at the same time to render all of the brightness differences of the object as density differences in the finished print. If the density in negative by which the highest light of the object is rendered be designated by the symbol D_{nmax} , and that negative density which renders the deepest shadow be designated by D_{nmin} , it is evident in order to fulfill the condition stated in the previous sentence that $D_{nmax} - D_{nmin} = ES_y$. The quantity $(D_{nmax} - D_{nmin}) = DS_n$ is termed the "density scale of the negative" and applies to a particular negative. This term (DS_n) should not be confused with the similar term DS_x which applies to the nega-

tive material itself with a meaning exactly parallel to that of DS_v , as applied to the positive material.

From a consideration of Fig. 10, it will be seen if the negative and printing paper be such that $DS_n = ES_v$, that all of the density differences in the negative will be rendered as density differences in the print, and further, that the maximum negative density will be rendered as the minimum useful density of the printing material while the minimum negative density will result in the maximum useful density in the print. Such procedure, of course, results in the loss of contrast in the extreme shadows and highlights due to the shape of the characteristic curve of the printing paper, even though (as is usually possible) all of the object brightnesses are rendered on the straight line portion of the negative curve. Of two printing papers having identical values of ES_v (and equal to the value of DS_n for the negatives considered), the better result will in general be obtained by use of the one of which the characteristic curve most closely approximates to a straight line between the points m and n . While in some special cases better results may be obtained by some other procedure, it is undoubtedly true that in general the most favorable relation between negative and printing paper is that the "density scale" DS_n of the former shall be as nearly as possible equal to the "exposure scale" DS_v of the latter.

Since it is not the primary object of this paper to deal with particular cases, but rather to outline the general theory of the subject, the discussion of the objective phase will be closed at this point, leaving the application to practical problems for treatment in subsequent papers. Moreover, a final evaluation of the quality of a reproduction cannot be made without a consideration of the subjective phase of the problem, and this will now be treated.

THE SUBJECTIVE PHASE.

Turning now to a consideration of the subjective phase, it will be well to discuss the significance of the curves shown in Fig. 3. These contain the data relative to the visual sensitivities necessary for the evaluation of the subjective relative-contrast function. Curve A gives the relation of the sensation to the stimulating brightness. Equal increments on the Y axis correspond to equal increments in the sensation and the brightness intervals which give rise to these sensation steps can be determined from this curve. The shape of this function could be established by

direct experiments, but since such a process is extremely laborious, it is customary to arrive at the same end by other means. The curve B , the contrast sensibility, is the first derivative of the sensation curve, hence, if the sensibility be measured and the curve B plotted, the sensation curve may be obtained by integration. The sensibility at a given adaptation level is determined by measuring the least difference in brightness perceptible under the specified conditions. The least perceptible difference, ΔB , expressed as a fraction of the brightness at which it is determined, is a direct measure of the sensibility of the eye to contrast or brightness difference. The data from which the curve B is plotted are given in Table IV, and are from the work of Nutting and Blanchard (*loc. cit.*). The ordinate of curve B at any point is therefore equal to the slope of the curve A at the same value of B . Thus, if S be the sensation, $\frac{dS}{d \log B} = \frac{B}{\Delta B}$. The values of the ordinates of the sensation curve are in terms of sensation units, and the construction of such a sensation scale is based on the assumption that a just perceptible difference at any point on the sensation scale is equal to a just perceptible difference at any other point. The basic unit upon which this scale is established is therefore the least perceptible brightness difference.

It will be noted from the data in Table IV that the range of brightness over which the eye operates is enormous, from $B = .000001$ to $B = 10000$ millilamberts, a range of 1 to 10,000,000,000, while for the photographic plate the corresponding range is but 1 to 2000 (approximately). This extreme range in case of the eye is due largely to the fact that the retina changes in sensibility with the intensity of the incident radiation. This property of variable sensibility has been likened to that of an ammeter equipped with a shunt whose resistance decreases as the current increases. The diameter of the pupil is but a very small factor in the sensibility change since it varies in area over a range of but 1 to 64. It should be pointed out that the curves in Fig. 3 are similar in shape to those given by Renwick (*loc. cit.*) but that the absolute values of log brightness are very different. This is due undoubtedly to the uncertainty in the value of the unit used by König, from whose data the curves in question were plotted. Blanchard (*loc. cit.*) has discussed the question and has computed the probable value of König's unit as being .004 ml. (approximate).

mately). The millilambert is the c. g. s. unit of brightness, and is equivalent to a brightness of ten meter candles or .93 foot candles. The region of maximum sensibility is therefore from 3 to 250 foot candles (approximately), instead of from 20,000 to 400,000 as indicated by the curves published by Renwick. Since the function of use in relating subjective reproduction to the objective contrast is obtained by using the ratio of the gradients at two points on the sensation curve, the units in which the curve is plotted are not of importance. However, in order to determine the sensibility for a given adaptation level, it is important that the correct abscissæ value be used.

Now, if we consider a specified brightness difference in the object viewed with the eye adapted to some brightness level A_0 , it is possible from curve A (Fig. 3) to find the corresponding sensation difference; and if the brightness difference between the corresponding areas of the material reproduction be known, together with the adaptation level (A_{mr}) of the observer viewing this material reproduction, it is possible to determine also the

TABLE IV.

Visual Sensitometric Data

Field Brightness	Difference Fraction	Discrimination Factor	Threshold Limit
0.000001	(1.00)	1.0	0.00000093
0.00001	(0.66)	1.5	0.0000042
0.0001	0.395	2.5	0.000019
0.001	0.204	4.5	0.000087
0.01	0.078	12.8	0.00039
0.1	0.0370	27.0	0.00174
1.0	0.0208	48.2	0.0081
10.0	0.0174	57.5	0.036
100.0	0.0172	58.1	0.28
1000.0	0.0240	41.7	2.15
10000.0	(0.048)	(20.9)	(232.0)

magnitude of the resulting subjective contrast. A comparison of the magnitudes of these two contrast values namely: (a) the subjective contrast resulting from observation of the object at a given adaptation level, which subjective evaluation it is convenient to refer to as the subjective object; and (b) the subjective contrast due to the observation of the material reproduction at a specified adaptation level, which subjective evaluation will be termed the

subjective reproduction; determines the exactness with which the subjective sensation resulting from the observation of the object is reproduced by observation of the material reproduction (the illuminated positive) under the conditions specified. It is also possible by similar methods to determine the magnitude of the contrast necessary, at any point in the tonal scale of the material reproduction, for the exact reproduction of the subjective contrast of the object, assuming, of course, that the values of A_o and A_{mr} are known

A_o = Adaptation level while observing object.

A_{mr} = Adaptation level while observing material reproduction.

For instance, in Fig. 12 consider an increment of brightness, $\Delta \log B_1 = \log B_1 - \log B_1'$, for which the corresponding increment in the sensation is $\Delta S_1 = (S_1 - S_1')$. At any other point take a second increment in sensation $\Delta S_2 = (S_2 - S_2')$ such that $\Delta S_1 = \Delta S_2$. Let the increment in $\log B$, corresponding to ΔS_2 , be represented by $\Delta \log B_2 = (\log B_2 - \log B_2')$. Assume that the increment $\Delta \log B_2$ applies to some point in the object, and that the value of $\log B_2$ fixes the adaptation level of the observer looking at that object, while for the material reproduction the adaptation is conditioned by the value of B_1

$$A_o = B_2$$

$$A_{mr} = B_1$$

Now the average slope of the curve (Fig. 12) over the range $\Delta \log B_2$ is given by,

$$\text{Slope (2)} = \frac{\Delta S_2}{\Delta \log B_2}.$$

And the average slope over the interval $\Delta \log B_1$ is likewise,

$$\text{Slope (1)} = \frac{\Delta S_1}{\Delta \log B_1}.$$

Since $\Delta S_1 = \Delta S_2$ it follows that

$$\frac{\text{Slope (2)}}{\text{Slope (1)}} = \frac{\Delta \log B_1}{\Delta \log B_2}$$

But the average slope of the sensation curve over any given interval is given by the average value of the ordinate, D , of the contrast sensibility curve (Curve B , Fig. 3) over the same in-

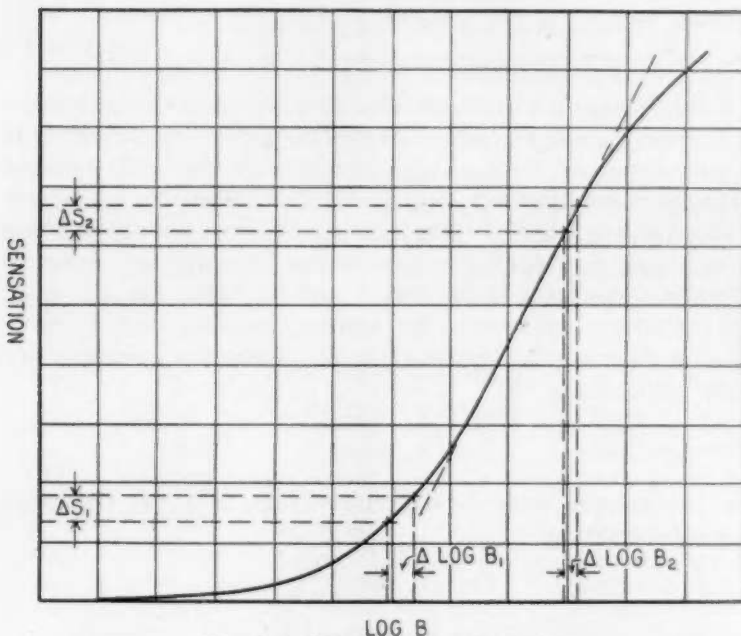
terval. Hence, we may express the values of the slope by the corresponding values of D , and equation becomes

$$\frac{\text{Average } D_2}{\text{Average } D_1} = \frac{\Delta \log B_1}{\Delta \log B_2}$$

or

$$\Delta \log B_2 = \Delta \log B_1 \cdot \frac{\text{Average } D_1}{\text{Average } D_2}$$

FIG. 12.



The relation between sensation and stimulus.

It is evident, therefore, if $\Delta \log B_2$ be the objective brightness difference at some point in the object that the equation will give the value of the $\Delta \log B_1$, the objective brightness difference necessary in the photographic reproduction for the exact reproduction of subjective contrast. In the case illustrated, the interval $\Delta \log B_2$ is less than the interval $\Delta \log B_1$, thus indicating that a greater objective contrast in the reproduction will be required to reproduce the subjective contrast due to the smaller objective contrast in the object.

It is well known that other factors than the brightness of the area upon which the attention is fixed (foveal image) influence to some extent the adaptation level of the eye at any instant. Such factors include the brightness of the surrounding objects (*i.e.*, the peripheral images), and the length of time during which the attention has been fixed on the area considered. In practice, however, the picture being observed occupies a very considerable portion of the field of vision, and, further, in the great majority of such pictures the actual range of brightness is relatively limited. This is especially true of surfaces viewed by reflected light. Further, under the majority of conditions, the field of vision not filled by the reproduction being considered does not contain any area contrasting extremely with the brightness of the reproduction. It seems reasonable, therefore, to assume that the adaptation level will be fixed in practice by the brightness of the reproduction itself, and likewise that the level when viewing the object will be conditioned by the object considered.

The question then arises as to what particular brightness in object and reproduction will determine the adaptation level in each case. Considering the reproduction as a reflecting surface, the average range of brightness may be taken as 1 to 40 (1.6 in $\log B_{mr}$), which, even at the steepest part of the sensibility curve (Fig. 3) does not correspond to an excessive change in value of D . The change in adaptation as attention travels from high-light to shadow probably is not so great as that indicated by the brightness range, being limited to some extent by the stabilizing effect of the peripheral images as well as to the time lag of adaptation behind the changing foveal stimulus. It seems logical, therefore, to assume that A_o and A_{mr} are conditioned by the average brightness of object and material reproduction, and this assumption will undoubtedly hold for a large percentage of the normal cases, although it may be necessary, under extreme conditions, to take into account the change in adaptation (and consequently in the value of D) which occurs as attention shifts from an area of one brightness to that of another.

As a simplifying condition, let us, therefore, assume that the adaptation level is conditioned by a brightness of the object and reproduction respectively half-way between (on the $\log B$ scale) the highest and lowest brightness considered.

Let the values of the various factors be designated by:

For the object:

A_o = Adaptation Level for Average B_o .

D_o = Sensibility.

For the reproduction:

A_{mr} = Adaptation Level for Average B_{mr} .

D_{mr} = Sensibility.

Now in the equation previously given,

$$\frac{\text{Average } D_1}{\text{Average } D_1} = \frac{\Delta \log B_1}{\Delta \log B_1}$$

let the numerical subscripts be replaced by the subscript letters indicating the application of the terms to particular values, thus,

$$\frac{\text{Average } D_o}{\text{Average } D_{mr}} = \frac{\Delta \log B_{mr}}{\Delta \log B_o}$$

By allowing these finite increments to approach zero as a limit, we may replace the average values of D by values of the slope of the contrast sensibility curve at a point and obtain the expression

$$\frac{D_o}{D_{mr}} = \frac{d \log B_{mr}}{d \log B_o}$$

Since previously an assumption of equality of subjective contrast (sensation increment) was made, this equation is a statement of the conditions necessary for the exact reproduction of the subjective contrast.

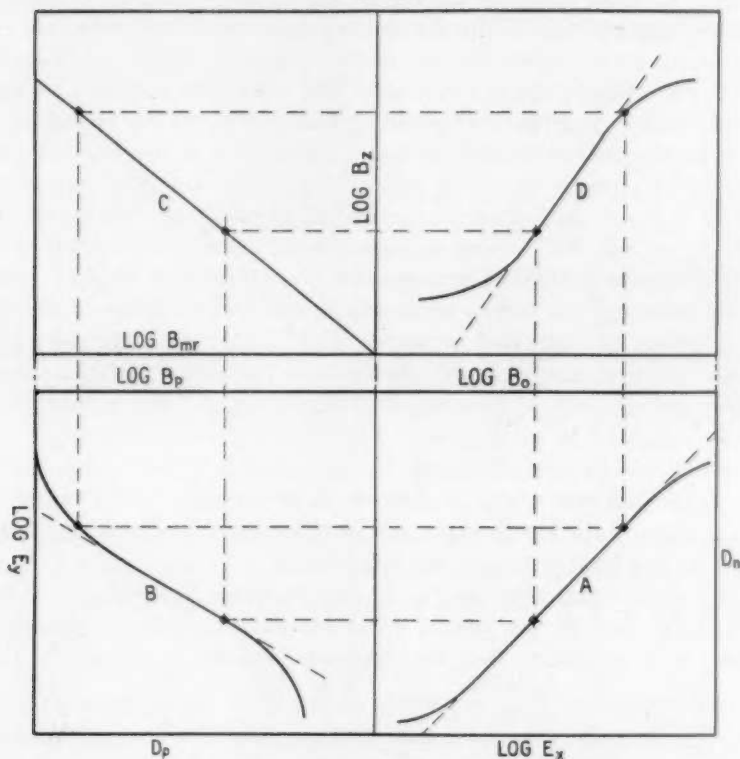
Now the value of $\frac{D_o}{D_{mr}}$ gives the slope of a curve which if plotted in Quadrant I, Fig. 4, would be the line of exact subjective reproduction of objective contrast. In the graphic solution, however, it is more convenient in plotting the reproduction curve D , and also in the later interpretation of the results, to retain the line E (of which the gradient is unity) as the line of exact contrast reproduction, and to alter the position of the points on the scale on the line O so that the reproduction curve plotted therefrom will be the curve of actual subjective reproduction, this alteration of position being an expansion or compression of the points on the scale in the case of constant values for the adaptation levels (A_o and A_{mr}), or a distortion of their distribution in case

such values are variable. It is quite possible to obtain the final evaluation of the exactness of reproduction as a comparison between the contrast values of the subjective object and the subjective reproduction (see Table I, Items I and X), but in view of the fact that the starting point in any tone reproduction problem must be the objective brightness values of the object, it seems more logical to make these values the primary base of the computation and to express finally the departures from exact reproduction in terms of these physically measured values of object brightness. This necessitates the evaluation of the subjective contrast of the reproduction in terms of the subjective contrast of the object, and a final expression of this complex relation as a function of the object brightness B_o . The final comparison must therefore be made between the actual measured (objective) contrast values of the object and the corresponding contrast values of the subjective reproduction evaluated in terms of the subjective object. The scale on which the points representing the various areas of the reproduction as evaluated in terms of this relative subjective contrast function, must occupy the line OY in order that the curve representing the final reproduction may be graphically constructed with values of B_o as abscissæ. The required scale constructed on line OY will be termed the $\log B_z$ scale, and the relative subjective contrast function which is used to determine the distribution of the points on the $\log B_z$ scale will be referred to as the Z function, and its gradient will be designated by G_z .

Now in order that the line E (in Quadrant I), having a slope of unity, shall be the line of exact subjective contrast reproduction, it is necessary that its slope or gradient be given by the expression $G_c = \frac{\Delta \log B_z}{\Delta \log B_o}$ where $\Delta \log B_o$ is some increment on the $\log B_o$ scale for which the corresponding interval in the subjective reproduction is $\Delta \log B_z$. Let the scale of $\log B_{mr}$ on line OY (Fig. 4) be replaced by a scale of $\log B_z$ (Fig. 13). This can be accomplished by replacing the dummy curve C (Fig. 4), by the new function which gives the relation between the material (objective) reproduction and the subjective reproduction. This subjective reproduction is itself a complex function containing the relation between the subjective evaluation on the object and subjective evaluation of the material reproduction. These evaluations are conditioned by the sensibility of the eye to brightness dif-

ferences (contrast) at the time of observing the object and when viewing the reproduction. This relation, the relative subjective contrast function, can be constructed graphically, but the consideration of this will be omitted. It is sufficient for the present purpose to state that the gradient of this curve, which must be

FIG. 13.



Special case of the complete solution including subjective phase.

substituted for the dummy curve thus far used, is equal to the reciprocal of the ratio of the values of D_o to D_{mr} . Let G_s indicate the gradient of the relative subjective contrast function which from this point on will be used instead of the dummy curve, having an arbitrary gradient equal to unity.

$$G_s = \frac{D_{mr}}{D_o}$$

This procedure also makes it possible to establish on the line OY the scale $\log B_z$ (Fig. 13), for which the points located by the usual construction methods will represent the relative subjective brightnesses corresponding to the various areas of the object considered.

Now if

$$\begin{aligned} D_o &= D_{mr} \\ G_z &= 1.0 = G_d \end{aligned}$$

This is the gradient thus far used in the construction of the curve C . It is evident, therefore, if average B_o and average B_{mr} are such that they fall between the limits (Fig. 3), $\log B = 0.5$ ($B = 3.2$ ml.) and $\log B = 2.2$ ($B = 166$ ml.) that the subjective reproduction of contrast will be identical with the material reproduction curves already discussed.

If, however, the brightness, B_o , of the object and the consequent adaptation level, A_o , of the observer viewing that object is relatively high, while the brightness of the material reproduction, B_{mr} (the illuminated positive), and the corresponding adaptation level, A_{mr} , is low, the subjective reproduction will be different from the objective reproduction. In this case the contrast sensitivity D_o is greater than D_{mr} and hence the gradient, G_z , of the relative subjective contrast curve is less than unity.

$$G_z = \frac{D_{mr}}{D_o}$$

The reproduction curve, D , constructed in the usual way will also have a decreased gradient. This indicates that under such conditions a loss of contrast occurs due to the subjective factors, and in order to obtain exact subjective reproduction it will be necessary to enhance the objective contrast in material reproduction, in order to compensate for the loss due to the subjective effect. It is evident from these considerations that a picture of a brilliantly light scene, which is to be viewed under relatively low illuminations, such as exist in interiors at night should be somewhat more contrasty than the scene itself. That is, best subjective reproduction will be obtained by a positive in which the actual objective contrast is somewhat enhanced. This condition may be stated symbolically thus:

If,

$$\begin{aligned} D_o &\text{ is greater than } D_{mr}, \\ G_z &\text{ is less than } 1.0. \end{aligned}$$

When G_z is less than unity there will be a loss of contrast due to the subjective factors.

On the other hand if,

$$\begin{aligned} D_o &\text{ is less than } D_{mr}, \\ G_z &\text{ is greater than } 1.0, \end{aligned}$$

and there will be an enhancement of contrast due to the subjective factors. When G_z is greater than unity the slope of the relative subjective contrast function (Curve C is Quadrant IV) is greater than unity, and the gradient of the resulting reproduction curve D (Quadrant I) is increased accordingly.

Now from a consideration of Fig. 13 it will be seen that:

$$\begin{aligned} G_n &= \frac{d D_n}{d \log E_x} = \frac{d D_n}{d \log B_o} \\ G_p &= \frac{d D'd}{d \log E_y} = - \frac{d D'd}{d D_n} = \frac{d \log B_{mr}}{d D_n} \\ G_n \cdot G_p &= \frac{d \log B_{mr}}{d \log B_o} \\ G_z \text{ (Curve C)} &= \frac{d \log B_z}{d \log B_{mr}} = \frac{D_{mr}}{D_o} \\ G_r \text{ (Curve D)} &= \frac{d \log B_z}{d \log B_o} \\ G_n \cdot G_p \cdot G_z &= \frac{d \log B_{mr}}{d \log B_o} \cdot \frac{d \log B_z}{d \log B_{mr}} \\ &= \frac{d \log B_z}{d \log B_o} = G_r \\ \therefore G_n \cdot G_p \cdot G_z &= G_r \end{aligned}$$

For the straight line portions of the curves, if such exist,

$$\gamma_n \gamma_p \cdot \gamma_z = \gamma_r$$

Therefore if any three of these four functions involved be known or are assumed, the fourth may be determined. The line E as before is the curve of exact reproduction of contrast, and the curve D is the reproduction obtained under the specified conditions. Deviations may be determined as in case of the solution for the objective phase, and the results will indicate the departure

from the exact reproduction of subjective contrast. It should be noted that this construction gives direct comparison between the objective values of brightness and the relative subjective equivalent in the reproduction. That is, the comparison is not between the subjective impression of the object and the subjective impression due to the reproduction, although this comparison can easily be made if such seems desirable. The procedure adopted is such that all of the subjective factors are introduced by use of the curve *C*, the relative subjective contrast function, such procedure in effect is the evaluation of the subjective reproduction in terms of the subjective contrast in the object itself.

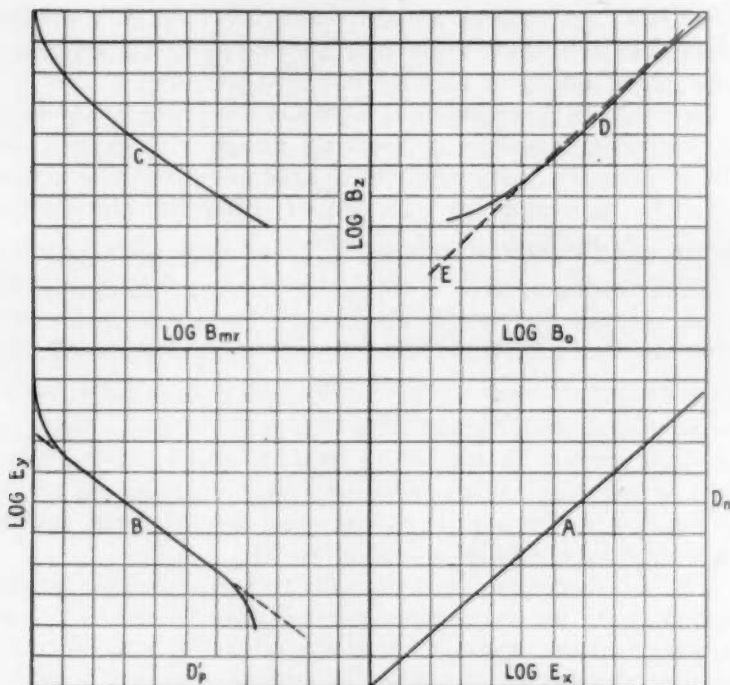
Now in case it is considered necessary, by reason of the existing conditions, to take into account the change of adaptation with shift of attention from highlight to shadow, this can be accomplished by computing for each pair of object and reproduction brightnesses the value of the ratio $\frac{D_{mr}}{D_o}$, and from the values thus obtained plotting a curve which is then used as the relative subjective contrast function. The curve *C* may not, under such circumstances, be a straight line, but will vary in slope as indicated by the computed values of $\frac{D_{mr}}{D_o}$. As a typical example of such a case computed from actual values of a practical case, curve *C* in Fig. 14 is given. This is for an object in which the brightness range is from 15 ml. to 1000 ml., while the intensity of the illumination on the positive during observation is such that the brightness range is from .35 ml. to 10 ml.

Fig. 14, therefore, represents the complete general graphic solution giving the reproduction curve *D*, which is a graphic representation of the reproduction of the subjective impression obtainable under any set of specified conditions, the line *E* being the curve of exact reproduction of subjective contrast. It should be noted that with the particular values assumed in this case, the reproduction curve *D* is somewhat straighter in the highlight region than when a straight line was used as the relative subjective contrast function. It is probable in practice that the change in adaptation level is not so great as is indicated by the assumptions made in this case, and that the improvement in contrast reproduction is not so great as indicated by the curve shown.

Many interesting relations between the various factors may be derived and applied to practical problems, but the discussion of

such details will not be taken up at this time. One point deserving mention at this time, however, is relative to the values of adaptation of the eye, upon which depend the sensibility values from which the slope of the subjective relative-contrast function is computed. Few data are at present available as to adaptation levels when the visual field is filled by areas of different bright-

FIG. 14.



A general case of the complete solution including the subjective phase.

nesses, such as exists under practical conditions. Experimental work is at present in progress in this laboratory, from the results of which it is hoped more reliable evaluations of A_o and A_{mr} may be obtained for certain specified practical conditions. When these data are available, it will be possible by the application of the general principles outlined in this paper to arrive at more certain conclusion, relative to the exactness of reproduction in any particular case.